WEIKAI LI

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EDUCATION

UCLA, Los Angeles, California (Ph.D. in Computer Science) 09/2023 – present

- GPA: 4.00/4.00
- Advisor: I am fortunate to be advised by Prof. Yizhou Sun. I also closely work under Prof. Jason Cong's guidance.
- Research areas: My recent research focuses on AI for chip design and AI models' interpretability, where the utilized AI models include Large Language Models (LLMs) and Graph Neural Networks (GNNs). Besides, I have broad research interests in LLM and GNN.

Tsinghua University, Beijing, China (B.Eng in Computer Science and Technology) 08/2019 – 06/2023

- GPA: 3.92/4.00 (ranking: 10/195)
- Outstanding Undergraduate Graduate of Tsinghua University (top 10%)
- Advisors: I was fortunate to be advised by Prof. Jie Tang, Prof. Yuxiao Dong, and Prof. Juanzi Li.
- Research areas: Graph Neural Network (GNN); Large Language Models (LLMs)' in-context learning.

RESEARCH EXPERIENCES

Graduate Student Researcher at UCLA 09/2023 – present

Advised by Prof. Yizhou Sun

- **Hierarchical MoE for HLS prediction**: Enhanced the domain generalizability of GNN models for high-level synthesis (HLS) prediction by incorporating hierarchical mixture-of-experts (MoE).
- **Micro-architecture-informed neural network for HLS prediction**: Designed a novel model by utilizing a hardware analytical model with a neural network, making it composable and generalizable.
- **LLMs' interpretability**: Systematically analyzed how LLM's post-training process influences an LLM's internal mechanisms.
- **I am also researching on various topics related to LLM, GNN, chip design, and interpretability:** agentic workflow for HLS design, cross-modality interpretation of GNN and LLM, LLM's cognitive science behaviors under context change, etc.

Undergraduate Student Researcher at Tsinghua University 01/2023 – 07/2023

Advised by Prof. Juanzi Li

- **LLMs' in-context learning**: Analyzed the limitations of LLMs' in-context learning on the event relation extraction task, and we demonstrated the importance of finetuning for specification-heavy tasks.

Undergraduate Student Research Intern at UCLA 07/2022 – 12/2022

Advised by Prof. Yizhou Sun

- **GNN's approximation**: Designed an efficient approximation method to calculate the influence of removing a node on a GNN model's prediction, leading to faster detection of influential nodes.

Undergraduate Student Researcher at Tsinghua University 08/2021 – 07/2022

Advised by Prof. Jie Tang and Prof. Yuxiao Dong

- **GNN benchmark**: Discovered the phenomenon of over-tuning hyper-parameters in GNN model evaluation, and we proposed a new benchmark to avoid this problem and improve evaluation stability.
- **GNN model design**: Used the mixture-of-experts (MoE) structure to increase GNN's parameter size and improve its performance on complex datasets.

INTERNSHIP EXPERIENCES

Applied Scientist Intern at Amazon 06/2025 – 09/2025

Graph Machine Learning Team; Manager: Xiang Song; Mentor: Han Xie

- **GNN's interpretability**: Proposed a novel framework to summarize GNNs' patterns with LLMs.

PUBLICATIONS

[1] Jade Xu, Renliang Sun, Zijian Ding, **Weikai Li**, Kaiqiao Han, Han Zhang, Chengyue Wang, Haixin Wang, Wei Wang, Yizhou Sun, Jason Cong. "How well do LLMs understand code?" Communications AI & Computing 2026.

[2] Mingfei Lu, Mengjia Wu, Feng Liu, Jiawei Xu, **Weikai Li**, Haoyang Wang, Zhengdong Hu, Ying Ding, Yizhou Sun, Jie Lu, Yi Zhang. "Choosing How to Remember: Adaptive Memory Structures for LLM Agents." Preprint, 2026.

[3] Ching Chang, Yijia Xiao, Jade Xu, Fred Xu, Chenchen Ye, Ruoyan Li, Junkai Zhang, Yihe Deng, Kyle Zheng, Ethan Ji, Alexander K Taylor, **Weikai Li**, Maryam Haghifam, Anthony Cuturrufo, Renliang Sun, Jiahang Sha, Yidan Shi, Wen-Chih Peng, Yizhou Sun, Wei Wang. "Training Recipes for Agentic Reinforcement Learning in LLMs: A Survey." Preprint, 2026.

[4] Deming Chen, Vijay Ganesh, **Weikai Li**, Yingyan Celine Lin, Yong Liu, Subhasish Mitra, David Z. Pan, Ruchir Puri, Jason Cong, Yizhou Sun. "Report for NSF Workshop on AI for Electronic Design Automation." IEEE CAS Magazine 2026.

[5] Mingfei Lu, Mengjia Wu, Jiawei Xu, **Weikai Li**, Feng Liu, Ying Ding, Yizhou Sun, Jie Lu, Yi Zhang. "From Newborn to Impact: Bias-Aware Citation Prediction." WWW 2026.

[6] Arvind Murari Vepa, Yannan Yu, Jingru Gan, Anthony Cuturrufo, **Weikai Li**, Wei Wang, Fabien Scalzo, Yizhou Sun. "A Multimodal LLM Approach for Visual Question Answering on Multiparametric 3D Brain MRI." KDD AI for Sciences Track 2026.

[7] Zijian Ding, Tung Nguyen, **Weikai Li**, Aditya Grover, Yizhou Sun, Jason Cong. "Iceberg: Enhancing HLS Modeling with Synthetic Data." LAD 2025.

[8] Hongzhe Du*, **Weikai Li***, Min Cai, Karim Saraipour, Zimin Zhang, Himabindu Lakkaraju, Yizhou Sun, and Shichang Zhang. "How Post-Training Reshapes LLMs: A Mechanistic View on Knowledge, Truthfulness, Refusal, and Confidence." COLM 2025; Outstanding Paper Award at the New England NLP workshop 2025.

[9] **Weikai Li**, Ding Wang, Zijian Ding, Atefeh Sohrabizadeh, Zongyue Qin, Jason Cong, and Yizhou Sun. "Hierarchical Mixture of Experts: Generalizable Learning for High-Level Synthesis." AAAI 2025.

[10] Zijian Ding, Atefeh Sohrabizadeh, **Weikai Li**, Zongyue Qin, Yizhou Sun, and Jason Cong. "Efficient Task Transfer for HLS DSE." ICCAD 2024.

[11] Yunsheng Bai, Atefeh Sohrabizadeh, Zijian Ding, Rongjian Liang, **Weikai Li**, Ding Wang, Haoxing Ren, Yizhou Sun, and Jason Cong. "Learning to Compare Hardware Designs for High-Level Synthesis." MLCAD 2024.

[12] **Weikai Li**, Zhiping Xiao, Xiao Luo, and Yizhou Sun. "Fast Inference of Removal-Based Node Influence." WWW 2024.

[13] Jifan Yu*, Xiaozhi Wang*, Shangqing Tu, Shulin Cao, Daniel Zhang-Li, Xin Lv, Hao Peng, Zijun Yao, Xiaohan Zhang, Hanming Li, Chunyang Li, Zheyuan Zhang, Yushi Bai, Yantao Liu, Amy Xin, Nianyi Lin, Kaifeng Yun, Linlu Gong, Jianhui Chen, Zhili Wu, Yunjia Qi, **Weikai Li**, Yong Guan, Kaisheng Zeng, Ji Qi, Hailong Jin, Jinxin Liu, Yu Gu, Yuan Yao, Ning Ding, Lei Hou, Zhiyuan Liu, Bin Xu, Jie Tang, and Juanzi Li. "KoLA: Carefully Benchmarking World Knowledge of Large Language Models." ICLR 2024.

[14] Hao Peng*, Xiaozhi Wang*, Jianhui Chen*, **Weikai Li**, Yunjia Qi, Zimu Wang, Zhili Wu, Kaisheng Zeng, Bin Xu, Lei Hou, and Juanzi Li. "When does In-context Learning Fall Short and Why? A Study on Specification-Heavy Tasks." ArXiv preprint, 2023.

[15] Ziang Li*, Ming Ding*, **Weikai Li**, Zihan Wang, Ziyu Zeng, Yukuo Cen, and Jie Tang. "Rethinking the Setting of Semi-supervised Learning on Graphs." IJCAI 2022.

(*Equal contributions)

SELECTED AWARDS AND HONORS

- Amazon Trainium Fellowship 2025
- Outstanding Undergraduate Graduate of Tsinghua University (top 10%) 2023
- Comprehensive Excellence Scholarship for junior year (top 10%) 2022
- Comprehensive Excellence Scholarship for sophomore year (top 10%) 2021

- Academic Excellence Scholarship for freshman year 2020
- Runner-up in Tsinghua Innovation Competition, industry and technology track, freshman group 2019
- First Prize in the National High School Mathematics League 2018

ACADEMIC SERVICES

- Organizer: NSF Agent for Chip Design Automation Workshop 2026
- Volunteer: NSF AI for EDA Workshop 2024
- Conference reviewer: ICLR 2024, ICML 2024, ACMMM 2024, NeurIPS 2024, LOG 2024, AAAI 2025, ICLR 2025, WWW 2025, ICML 2025, ICCV 2025, NeurIPS 2025, COLM 2025's workshop, LOG 2025, ICLR 2026, WWW 2026, ICML 2026, NSF Agent for Chip Design Automation Workshop 2026，AAAI 2027, ICLR 2027
- Journal reviewer: IEEE TBD 2024, PLOS ONE 2024, IEEE TBD 2026

TEACHING EXPERIENCE

- 2024 fall, 2025 winter, and 2025 spring: TA of CS 31 (Introduction to Computer Science I), UCLA
- 2024 summer: TA of CS 97 (Introduction to Data Science), UCLA summer school for high school students

SKILLS

Coding skills

- Programming languages: Python, C, C++, Java, JavaScript
- Hardware description languages: VHDL, Verilog
- Others: HTML, CSS

Language skills

- Native in Chinese (mandarin)
- Proficient in English